

Wideband Covariance Magnetometry below the Diffraction Limit

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We experimentally demonstrate a method for measuring correlations of wideband magnetic signals with spatial resolution below the optical diffraction limit. Our technique employs two nitrogen-vacancy (NV) centers in diamond as nanoscale magnetometers, spectrally resolved by inhomogeneous optical transitions. Using high-fidelity optical readout and long spin coherence time, we probe correlated megahertz-range noise with sensitivity of $15 \text{ nTHz}^{-1/4}$. In addition, we use this system for correlated T_1 relaxometry, enabling correlation measurements of gigahertz-range noise. Under such externally applied noise, while individual NV centers exhibit featureless relaxation, their correlation displays rich coherent and incoherent dynamics reminiscent of superradiance physics. This capability to probe high-frequency correlations provides a powerful tool for investigating a variety of condensed-matter phenomena characterized by nonlocal correlations.

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Two-point spatiotemporal correlations play an important role in characterizing many physical systems, including nonclassical electromagnetic fields [1], macroscopic quantum coherence in superconductors and Bose-Einstein condensates [2–5], quantum fluctuations of spin ensembles [6,7], and magnetic orderings in quantum simulators [8–12] and materials [13–29]. In particular, the ability to quantify magnetic correlation could significantly advance understanding of nonlocal collective phenomena in two-dimensional (2D) systems, such as novel magnetic order [30–32], hydrodynamic transport [33,34], and superconductivity [35]. However, accessing the properties of collective magnetic excitations in realistic materials typically requires probing spectrally broadband correlations at nanometer length scales, which is challenging. Examples include kilohertz- to megahertz-range longitudinal and gigahertz-range transverse spin fluctuations in 2D magnets [34,36], dc- to gigahertz-range noise from stimulated phonon emission in graphene [37], current fluctuations [38–40], and Berezinskii-Kosterlitz-Thouless vortex unbinding in 2D superconductors [41].

Several approaches have been explored to probe spin-spin correlations with nanoscale resolution. For example, neutron [13–16] and x-ray scattering [17], magnetostriction measurements [18,19], and nuclear magnetic-quadrupole resonance [20–22] can be used, albeit only for macroscopic, homogeneous samples. Optical techniques [23–29] are suitable for studying two-dimensional and thin-film

materials, but diffraction typically limits the resolution to a fraction of optical wavelength. Moreover, with the exception of optical pump-probe spectroscopy [28,29], none of the techniques above are suitable to probe broadband (dc to gigahertz) dynamics due to various technical limitations [19,20,24,26,42].

Atomlike systems such as nitrogen-vacancy (NV) centers in diamond offer a promising approach to this challenging problem. Such solid-state spin defect magnetometers have been used in “single-point” modality (i.e., probing a magnetic field at each point by a single defect or averaging over multiple defects in one optical spot) to study a wide range of static and dynamical phenomena in materials [34,43–49]. Recently, Rovny *et al.* [50] demonstrated that this system can be used to measure two-point spatiotemporal magnetic correlations. This approach has been extended to parallel measurement of multiple single NVs in different optical spots [51,52] or two NVs in the same spot [53,54]. These methods have been used to probe the magnetic noise up to megahertz frequency range, but they cannot be directly extended to gigahertz noise, which ubiquitously appears in various condensed-matter systems [34,36–40,48,49,55–63].

In this Letter, we extend covariance magnetometry to both new spatial and frequency scales by measuring magnetic field correlations up to gigahertz frequencies below the diffraction limit. Utilizing optical superresolution techniques to address individual NV centers based on inhomogeneous optical transitions [64–67], together with resonantly assisted spin-to-charge conversion [68], we achieve low readout noise, resulting in large detectable

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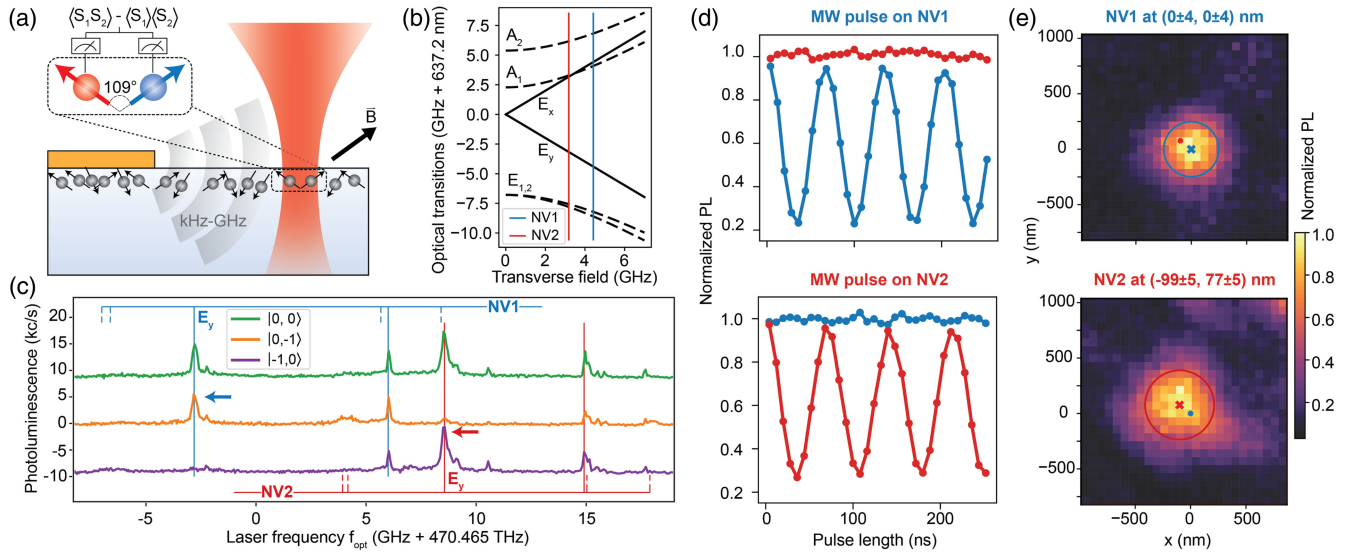


FIG. 1. Superresolution and independent spin control. (a) Schematics of the experimental system, consisting of two NV centers in a single optical spot experiencing a common magnetic field. The field is probed by measuring spin-spin correlation of the two NVs. NV1 (blue) and NV2 (red, color code used throughout the Letter) align differently to a static magnetic field of 10 mT. Both the MW signals driving spin transitions and test signals producing correlated magnetic fields are delivered through a gold stripline (yellow rectangle) fabricated on the diamond. (b) Calculated optical transitions of an NV as a function of transverse strain and electric field. The annotations, used as names of each transition, are the destination orbital states in the excited-state manifold. Vertical lines mark each NV's transverse field obtained from (c). Solid (dashed) black curves in (b) and vertical lines in (c) are $m_s = 0 (\pm 1)$ transitions. (c) PLE spectroscopy, utilizing different initial spin-state combinations of two NVs (three spectra, offset for clarity), enables identifying each NV's set of transitions. Arrows indicate transitions used for readout. (d) Independent spin control confirmed by driving Rabi oscillations of one NV and reading both defects. (e) Confocal maps taken by resonantly exciting NV1 (top) and NV2 (bottom). The location of the on-resonant NV (cross) is determined from a fit to a 2D Gaussian (full width at half maximum denoted by a circle). The off-resonant NV is marked by a smaller dot. All experiments in this Letter are done in a cryostat at 11 K.

correlations. We use this system to probe a correlated megahertz-range magnetic noise by a Ramsey-based protocol. In addition, we demonstrate that corelaxation of two NVs enables detailed studies of gigahertz-range signals. In particular, by externally applying such magnetic noise with correlated amplitudes, we observe superradiantlike dynamics—both coherent and incoherent—arising from local detuning and dephasing. This technique can be integrated with other state-of-the-art NV sensing platforms, such as scanning tips and nanopillar arrays [69], while scaling favorably to few-NV clusters. It is therefore particularly suited for probing two-point and higher-order magnetic correlations in condensed matter at different length scales (from nano- to micrometers) and across a broad spectrum (from dc to gigahertz).

Experimental system—Our experiments make use of two shallowly implanted NV centers in the same optical spot addressed by a single-path scanning confocal microscope [Fig. 1(a)] (for sample details, see Supplemental Material [70]). We choose a pair of NVs with different crystallographic orientations, resulting in distinct microwave (MW) transitions that enable independent spin control. While recent work has used global excitation schemes that require multiple readout phase combinations or advanced data analysis [53,54], we independently address each NV by

a defect-selective, spin-selective optical excitation [64–67]. This is possible because, at cryogenic temperatures, the spin-dependent ground-to-excited-state transitions are spectrally resolvable, and the inhomogeneous local strain and electric field separates each NV's set of optical transitions [79] [Fig. 1(b)]. By preparing the NVs in different spin-state combinations and performing photoluminescence excitation (PLE) spectroscopy, we identify the set of transitions for each NV [Fig. 1(c)]. We use the bright, high spin contrast $m_s = 0$, E_y transition of both NVs for readout in all following experiments.

To confirm independent spin control, we drive Rabi oscillations of one NV and read both defects sequentially in the same repetition. Figure 1(d) shows high spin contrast (75%–80%) for the driven NV and virtually no change in the undriven one. Superresolution imaging of NVs is also straightforward in this modality and is accomplished by performing two confocal scans, each with the laser on resonant with one NV [Fig. 1(e)]. The measured 125 nm lateral distance and estimated ~ 20 nm z distance [70] imply negligible magnetic dipolar interactions between the two NVs.

High-fidelity readout and correlation measurements—Our goal is to measure the shot-to-shot spin-spin correlation of two NVs, quantified by the Pearson correlation

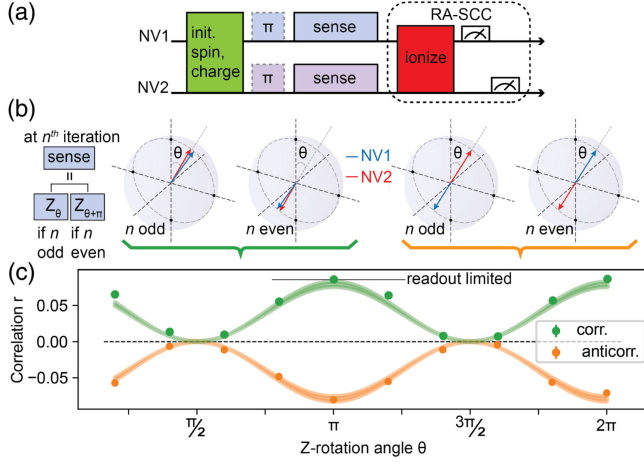


FIG. 2. Maximum detectable correlation. (a) General protocol for correlation measurements: after initialization to the $(\text{NV}^-, |0\rangle)$ state by 532 nm light, an optional π pulse prepares the spin state for sensing. High-fidelity readout is achieved by RA-SCC (see Appendix). (b) Demonstration of correlation measurement by direct NV spin driving, alternating between polar angle θ or $\theta + \pi$. (c) Pearson correlation coefficient r for both correlation (same initial states, green) and anticorrelation (opposite initial states, orange) configurations. Filled curves are theory predictions; the curve width denotes uncertainty of measured σ_R .

coefficient [50]

$$r = \frac{\langle S_1(r_1, t_1) S_2(r_2, t_2) \rangle - \langle S_1(r_1, t_1) \rangle \langle S_2(r_2, t_2) \rangle}{(\sigma_{p1} \sigma_{p2})(\sigma_{R1} \sigma_{R2})}, \quad (1)$$

where i is the NV index, $S_i(r_i, t_i) \in \{0, 1\}$ represents the spin state at time t_i , σ_{pi} is the quantum projection noise, and σ_{Ri} is the single-shot readout noise. $S = 0$ (1) corresponds to the $|m_s = 0\rangle$ (-1) state; $\langle \dots \rangle$ denotes averaging over all experimental repetitions. Since r depends on $1/\sigma_R^2$, it is imperative to have low σ_R . Therefore, we implement resonantly assisted spin-to-charge conversion (RA-SCC) [68] and achieve $\sigma_R = 3 - 4$ for both defects (see Appendix).

The overall protocol for all correlation measurements is as follows [Fig. 2(a)]: initialize each NV charge and spin state to $(\text{NV}^-, |0\rangle)$ by 532 nm light, prepare them in $|0\rangle$ or $|-1\rangle$ depending on the specific experiment, apply a MW sensing sequence to each NV independently, and do RA-SCC with sequential charge readout of each NV.

We benchmark our protocol by measuring correlations arising from direct NV spin driving [Fig. 2(b), [51,52]]. In the correlation configuration (green), both defects are prepared in the same spin state, alternating between either polar angle θ or $\theta + \pi$ on the Bloch sphere in each iteration. The measured correlation in Fig. 2(c) agrees well with the theory prediction $r = \cos^2(\theta)/(\sigma_{R1} \sigma_{R2})$ [70]. Intuitively, although alternating between θ and $\theta + \pi$ cancels the single-NV quantum projection noise (QPN) and makes

$\langle S_i \rangle = \frac{1}{2}$ regardless of θ , it does not cancel the QPN of the product $S_1 S_2$, thus $\langle S_1 S_2 \rangle$ remains θ dependent [70]. Therefore, r is maximized when $\theta = n\pi$ for integer n due to no QPN, and $r \approx 0$ for $\theta = (n + \frac{1}{2})\pi$ because the maximum QPN destroys any existing correlation. We also measure in the anticorrelation configuration (orange), in which both defects are prepared in opposite states, and observe the same results but with opposite sign of r , as expected. All reported correlation values hereafter are obtained after subtracting the small background correlation common to both correlation and anticorrelation configurations. Because the NVs undergo no free evolution in this experiment, the effects of spin decoherence are minimized, which enables quantifying the maximum detectable correlation $r_{\max} = 1/(\sigma_{R1} \sigma_{R2})$. The measured $r_{\max} = 0.083(3)$ agrees well with $r_{\max}^{\text{calc}} = 0.079(4)$ calculated from the measured σ_{R1} , σ_{R2} in Fig. 5(c).

Correlated T_2 spectroscopy—Next, we demonstrate sensing correlation of a $f_0 = 2.5$ MHz magnetic field phase modulated by a Gaussian noise with 25 kHz bandwidth. The XY8- N sensing sequence of each NV can be tuned independently with interpulse spacing τ_i to probe frequency $1/(2\tau_i)$, number of repetitions $N_i \in \{5, 8\}$, and relative delay Δt between each NV's sequence [Fig. 3(a)].

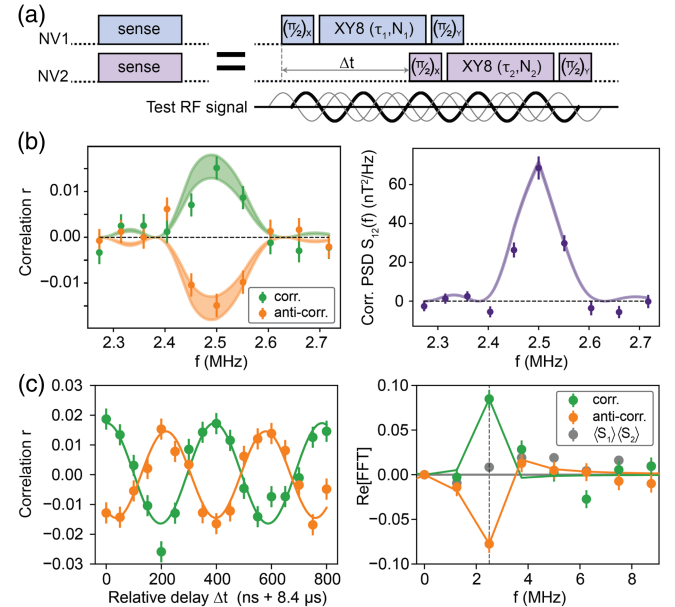


FIG. 3. Correlated T_2 spectroscopy. (a) The sensing sequence with phase-noisy ac test signal. (b) Equal-time ($\Delta t = 0$) spectroscopy, sweeping $\tau_1 = \tau_2 = \tau$ to probe $f = 1/(2\tau)$. PSD of the correlated noise (right) as calculated from the correlations (left). Filled curves represent theory fits, with a global scaling as the sole fit parameter; the curve widths indicate the uncertainty in the measured values of σ_R , B_1 , and B_2 [70]. (c) Two-time spectroscopy, sweeping Δt while keeping $\tau_1 = \tau_2 = 1/(2f_0)$, showing correlations with sinusoidal fits (left) and the real part of the corresponding Fourier transforms (right). Gray data points denote $\langle S_1 \rangle \langle S_2 \rangle$; vertical dashed line marks f_0 .

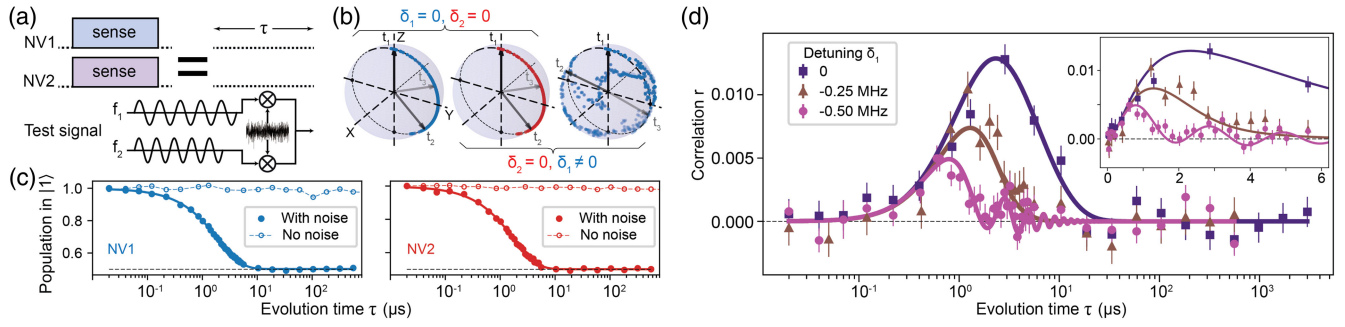


FIG. 4. Correlated T_1 spectroscopy. (a) The test signal is generated by combining two MW signals, each resonant with one NV center and independently tuned to have similar powers. Their amplitudes are modulated by the same 10 MHz bandwidth Gaussian noise. (b) Illustration of dynamics: With no detunings for both MW signals, the two NVs exhibit stochastic yet identical dynamics (first two panels), resulting in high correlation. Three Bloch vectors represent the state of the NVs at arbitrary times $t_1 < t_2 < t_3$. When the test signal for NV1 is detuned (third panel), despite the noise profile being identical to that of NV2 (second panel), NV1's trajectory includes precession around the z axis. As such, the correlation diminishes and oscillates at the precession rate. The fading color indicates later times. (c) Spin polarization relaxation of each NV under the test noise overlaid with a simple exponential decay, where $T_{1,\text{NV1}} = 1.96(2)$ and $T_{1,\text{NV2}} = 1.77(3)$ μ s. (d) Pearson correlation r for different detunings δ_1 (with $\delta_2 = 0$ fixed), showing correlation buildup despite featureless single-NV relaxation observed in (c). Inset: enlargement of the first 6 μ s, highlighting oscillations of correlation at the largest detuning $\delta_1 = -0.5$ MHz. Solid curves are global fits of the entire dataset using a model based on the quantum master equation [Eq. (3)].

We first detect the signal spectroscopically by equal-time correlation measurement ($\Delta t = 0$), sweeping $\tau_1 = \tau_2$. Figure 3(b) (left) shows agreement between experimental data and calculated correlation line shape based on the measured σ_{R1} , σ_{R2} , and ac field amplitudes at each NV $B_1 = 1.944(11)$ μ T and $B_2 = 0.56B_1$ (due to different orientations), as obtained from single-NV magnetometry [70]. The power spectral density (PSD) of the correlated noise in Fig. 3(b) (right) shows that, despite sensing for only 8 μ s in each shot, much shorter than $T_2 > 1$ ms, and 40 min in total for each data point, we can detect noise strength of ~ 70 nT²/Hz with uncertainty of 4 nT²/Hz, corresponding to magnetic sensitivity of 15 nTHz^{-1/4} [70]. Extending sensing to 1 ms, as allowed by the long T_2 , with negligible cost to total experiment time (due to long readout) could enable detection of correlated noise strength < 1 nT²/Hz [70]. To measure two-time correlation, we sweep the relative delay Δt at fixed $\tau_1 = \tau_2 = 1/(2f_0)$, observing the oscillation of r at f_0 as expected, while $\langle S_1 \rangle \langle S_2 \rangle$ remains featureless because the noise has zero mean [Fig. 3(c)].

Correlated T_1 spectroscopy—We now extend the technique to probing correlated few-gigahertz noise, which is beyond the scope of T_2 spectroscopy due to the impractically high Rabi frequencies required. To this end, we apply a test noise such that each NV is subject to a magnetic field $\mathbf{B}_i(t) \propto G(t) \cos(\omega_i t) \hat{x}$, where ω_i is close to the i th NV's $|0\rangle \rightarrow |-1\rangle$ transition Δ_i with controllable detuning $\delta_i \equiv \Delta_i - \omega_i$, and $G(t)$ describes a Gaussian noise with zero mean and bandwidth $\nu_c \simeq 10$ MHz. Such noise, common to both sensors and mimicking gigahertz magnetic field correlations, represents amplitude-modulated but phase-locked fluctuations [Fig. 4(a)].

While this noise causes the mean spin polarizations to experience simple exponential decays [Fig. 4(c)], the

correlation r undergoes richer dynamics [Fig. 4(d)]: initially, it develops as the two NVs corelax; next, application of a finite local detuning $\delta_1 \neq 0$ on NV1 results in oscillations of r at δ_1 ; finally, r decays to zero at longer times. Notably, the observed buildup, oscillations, and long-time decay of spin-spin correlation strikingly resemble the dynamics of dipole-dipole correlations in atomic and spin ensembles undergoing superradiance [80–83].

We explain these experimental observations by the following theoretical model. The effective Hamiltonian in corotating frames of each NV reads [70]

$$H = \sum_{i=1,2} \delta_i S_i^z + 2\mathcal{A}_i G(t) S_i^x, \quad (2)$$

where $S_i^{\alpha=x,y,z}$ are the spin-1/2 operators and $\mathcal{A}_i$ encodes the test signal amplitude. Provided that the bandwidth ν_c is much larger than other relevant dynamical scales, the noise is effectively Markovian and the time evolution of the two-NV density matrix $\rho(t)$ is governed by the quantum master equation (QME)

$$\partial_t \rho = i \sum_i \delta_i [S_i^z, \rho] + 2[\mathcal{L}, [\rho, \mathcal{L}]], \quad (3)$$

where $[A, B] \equiv AB - BA$ denotes the commutator and $\mathcal{L} \equiv (2\pi)^{1/4} \sqrt{1/\nu_c} (\mathcal{A}_1 S_1^x + \mathcal{A}_2 S_2^x)$ is the jump operator.

Equation (3) shows that $\langle S_i^z \rangle = 1/2 - \langle S_i^z \rangle$ relaxes as $\exp(-t/T_{1,\text{NV}i})$, in agreement with the data in Fig. 4(c). The extracted relaxation times $T_{1,\text{NV}i}$ correspond to field amplitudes $B_1 = 5.72(3)$ and $B_2 = 6.02(5)$ μ T.

To explain the dynamics of the correlation r , we note that, when each MW signal is resonant with the respective NV, i.e., $\delta_1 = \delta_2 = 0$, as follows from Eq. (2), both NVs are

stochastically driven around the x axis, albeit with identical trajectories that remain in the y - z plane [Fig. 4(b), first two panels]. The correlation is thus expected to develop over time and remain high. When the test signal is detuned for the first NV, i.e., $\delta_1 \neq 0$ and $\delta_2 = 0$, the correlation dynamics become more complex as NV1 undergoes both stochastic rotations around the x axis and precession around the z axis at δ_1 [Fig. 4(b), last two panels]. In this case, Eq. (3) predicts oscillations of the correlation at δ_1 . The observed decay at longer times originates from local and correlated dephasing effects, which we introduce phenomenologically [70]. With five fitting parameters for the entire dataset, this QME model closely matches the experimental data [Fig. 4(d)].

The analogy to superradiance can now be understood by looking at the structure of the jump operator $\mathcal{L}$ in Eq. (3), which is a sum of two terms that describe a common decay channel. In atomic superradiance, such a common decay channel originates from different atoms coupling to the same photonic modes, whose wavelengths significantly exceed the atom-atom spacings. In our system, this correlated decay is deliberately engineered via the common noise and only evident in the rotating frames of each NV center.

Conclusion and outlook—We have demonstrated covariance magnetometry with spatial resolution below the diffraction limit and probed correlated signals in both the mega- and gigahertz frequency bands. The combined effect of high-fidelity readout and long coherence time enables detection of correlated noise with magnetic sensitivity of $15 \text{ nTHz}^{-1/4}$, corresponding to detectable noise strength of $4 \text{ nT}^2/\text{Hz}$ in 40 min of total experimental time. This noise strength is well below that of critical fluctuations observed in several 2D magnets and superconductors [34,40,61,62] and is comparable to magnon density fluctuations in CrCl_3 [34]. With correlated T_1 spectroscopy, we observe correlation of two detuned gigahertz signals modulated by a common noise profile. Notably, the engineered magnetic field amplitudes are comparable to those reported for NV centers proximal to a magnon bath [49,84]. The observed superradiant-type dynamics of this correlation capture the coherent effect of local detuning and incoherent effect of dephasing. Combining our correlated T_1 protocol with dynamical decoupling schemes could be an interesting avenue for future work.

Our technique can be naturally extended in both the spatial and spectral domains, such as applying to few-NV clusters [65,67] in scanning-NV platforms [53,85] or incorporating techniques well developed for single-point NV sensing such as frequency conversion [86] and high-resolution spectroscopy [87,88]. Tuning the static field up to $\sim 100 \text{ mT}$ or implementing spin-locking relaxometry [89,90] should enable broad frequency tunability of the correlated T_1 protocol down to $\sim 100 \text{ MHz}$, a regime that can also be approached from below by the correlated T_2

protocol. Sensitivity could also be improved with various methods, from increasing photon collection efficiency with diamond patterning [85,91] to utilizing dipolar interactions in NV ensembles [92]. This Letter demonstrates that NV centers constitute a powerful nanoscale wideband probe of two-point and higher-order magnetic correlations, particularly suited to studying condensed-matter systems with intriguing correlated dynamics [93], such as superconductors [38–41], critical fluctuations in 2D magnets [34,60–62], nonequilibrium [37,94] and correlated transport phenomena [47,95], and nonlinear magnonic processes [34,36,49,55–59,96].

Note added—Recently, we became aware of related work demonstrating superresolution covariance magnetometry [97], which reports correlation measurements of signals with frequency up to a few megahertz in two NVs within a single optical spot by complementary detection methods, including utilizing an entangled pair of NV centers.

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Data availability—The data that support the findings of this article are openly available [98].

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- [1] M. O. Scully and M. S. Zubairy, *Quantum Optics* (Cambridge University Press, Cambridge, England, 1997).
 - [2] M. Tinkham, *Introduction to Superconductivity*, 2nd ed. (Dover Publications, Mineola, NY, 2004).
 - [3] M. R. Andrews, C. G. Townsend, H.-J. Miesner, D. S. Durfee, D. M. Kurn, and W. Ketterle, Observation of interference between two Bose condensates, *Science* **275**, 637 (1997).
 - [4] E. Altman, E. Demler, and M. D. Lukin, Probing many-body states of ultracold atoms via noise correlations, *Phys. Rev. A* **70**, 013603 (2004).
 - [5] S. Fölling, F. Gerbier, A. Widera, O. Mandel, T. Gericke, and I. Bloch, Spatial quantum noise interferometry in expanding ultracold atom clouds, *Nature (London)* **434**, 481 (2005).

- [6] J. L. Sørensen, J. Hald, and E. S. Polzik, Quantum noise of an atomic spin polarization measurement, *Phys. Rev. Lett.* **80**, 3487 (1998).
- [7] S. A. Crooker, D. G. Rickel, A. V. Balatsky, and D. L. Smith, Spectroscopy of spontaneous spin noise as a probe of spin dynamics and magnetic resonance, *Nature (London)* **431**, 49 (2004).
- [8] J. Simon, W. S. Bakr, R. Ma, M. E. Tai, P. M. Preiss, and M. Greiner, Quantum simulation of antiferromagnetic spin chains in an optical lattice, *Nature (London)* **472**, 307 (2011).
- [9] T. Fukuhara, P. Schauß, M. Endres, S. Hild, M. Cheneau, I. Bloch, and C. Gross, Microscopic observation of magnon bound states and their dynamics, *Nature (London)* **502**, 76 (2013).
- [10] M. F. Parsons, A. Mazurenko, C. S. Chiu, G. Ji, D. Greif, and M. Greiner, Site-resolved measurement of the spin-correlation function in the Fermi-Hubbard model, *Science* **353**, 1253 (2016).
- [11] T. Manovitz, S. H. Li, S. Ebadi, R. Samajdar, A. A. Geim, S. J. Evered, D. Bluvstein, H. Zhou, N. U. Köylüoğlu, J. Feldmeier, P. E. Dolgirev, N. Maskara, M. Kalinowski, S. Sachdev, D. A. Huse, M. Greiner, V. Vuletić, and M. D. Lukin, Quantum coarsening and collective dynamics on a programmable simulator, *Nature (London)* **638**, 86 (2025).
- [12] M. Bernaschi, I. González-Adalid Pemartín, V. Martín-Mayor, and G. Parisi, The quantum transition of the two-dimensional Ising spin glass, *Nature (London)* **631**, 749 (2024).
- [13] W. Marshall and R. D. Lowde, Magnetic correlations and neutron scattering, *Rep. Prog. Phys.* **31**, 705 (1968).
- [14] Y. Endoh, K. Yamada, R. J. Birgeneau, D. R. Gabbe, H. P. Jenssen, M. A. Kastner, C. J. Peters, P. J. Picone, T. R. Thurston, J. M. Tranquada, G. Shirane, Y. Hidaka, M. Oda, Y. Enomoto, M. Suzuki, and T. Murakami, Static and dynamic spin correlations in pure and doped La_2CuO_4 , *Phys. Rev. B* **37**, 7443 (1988).
- [15] T.-H. Han, J. S. Helton, S. Chu, D. G. Nocera, J. A. Rodriguez-Rivera, C. Broholm, and Y. S. Lee, Fractionalized excitations in the spin-liquid state of a kagome-lattice antiferromagnet, *Nature (London)* **492**, 406 (2012).
- [16] A. Rybakov, C. Boix-Constant, D. Alba Venero, H. S. J. van der Zant, S. Mañas-Valero, and E. Coronado, Probing short-range correlations in the van der Waals magnet CrSBr by small-angle neutron scattering, *Small Sci.* **4**, 2400244 (2024).
- [17] Y. Ishiguro, K. Kimura, S. Nakatsuji, S. Tsutsui, A. Q. R. Baron, T. Kimura, and Y. Wakabayashi, Dynamical spin-orbital correlation in the frustrated magnet $\text{Ba}_3\text{CuSb}_2\text{O}_9$, *Nat. Commun.* **4**, 2022 (2013).
- [18] E. Callen and H. B. Callen, Magnetostriction, forced magnetostriction, and anomalous thermal expansion in ferromagnets, *Phys. Rev.* **139**, A455 (1965).
- [19] V. S. Zapf, V. F. Correa, P. Sengupta, C. D. Batista, M. Tsukamoto, N. Kawashima, P. Egan, C. Pantea, A. Migliori, J. B. Betts, M. Jaime, and A. Paduan-Filho, Direct measurement of spin correlations using magnetostriction, *Phys. Rev. B* **77**, 020404 (2008).
- [20] T. Imai, C. P. Slichter, K. Yoshimura, and K. Kosuge, Low frequency spin dynamics in undoped and Sr-doped La_2CuO_4 , *Phys. Rev. Lett.* **70**, 1002 (1993).
- [21] P. Carretta, T. Ciabattini, A. Cuccoli, E. Mognaschi, A. Rigamonti, V. Tognetti, and P. Verrucchi, Spin dynamics and magnetic correlation length in two-dimensional quantum heisenberg antiferromagnets, *Phys. Rev. Lett.* **84**, 366 (2000).
- [22] K. Kitagawa, T. Takayama, Y. Matsumoto, A. Kato, R. Takano, Y. Kishimoto, S. Bette, R. Dinnebier, G. Jackeli, and H. Takagi, A spin-orbital-entangled quantum liquid on a honeycomb lattice, *Nature (London)* **554**, 341 (2018).
- [23] E. M. Haines, R. Clauberg, and R. Feder, Short-range magnetic order near the Curie temperature iron from spin-resolved photoemission, *Phys. Rev. Lett.* **54**, 932 (1985).
- [24] A. Ron, E. Zoghlin, L. Balents, S. D. Wilson, and D. Hsieh, Dimensional crossover in a layered ferromagnet detected by spin correlation driven distortions, *Nat. Commun.* **10**, 1654 (2019).
- [25] C. H. Back, C. Wüsch, A. Vaterlaus, U. Ramsperger, U. Maier, and D. Pescia, Experimental confirmation of universality for a phase transition in two dimensions, *Nature (London)* **378**, 597 (1995).
- [26] C. Jin, Z. Tao, K. Kang, K. Watanabe, T. Taniguchi, K. F. Mak, and J. Shan, Imaging and control of critical fluctuations in two-dimensional magnets, *Nat. Mater.* **19**, 1290 (2020).
- [27] L. Yang, P. Glasenapp, A. Greilich, D. Reuter, A. D. Wieck, D. R. Yakovlev, M. Bayer, and S. A. Crooker, Two-colour spin noise spectroscopy and fluctuation correlations reveal homogeneous linewidths within quantum-dot ensembles, *Nat. Commun.* **5**, 4949 (2014).
- [28] G. De Filippis, V. Cataudella, E. A. Nowadnick, T. P. Devereaux, A. S. Mishchenko, and N. Nagaosa, Quantum dynamics of the Hubbard-Holstein model in equilibrium and nonequilibrium: Application to pump-probe phenomena, *Phys. Rev. Lett.* **109**, 176402 (2012).
- [29] A. Sriram and M. Claassen, Light-induced control of magnetic phases in Kitaev quantum magnets, *Phys. Rev. Res.* **4**, L032036 (2022).
- [30] Q. H. Wang *et al.*, The magnetic genome of two-dimensional van der Waals materials, *ACS Nano* **16**, 6960 (2022).
- [31] L. Ciorciaro, T. Smoleński, I. Morera, N. Kiper, S. Hiestand, M. Kroner, Y. Zhang, K. Watanabe, T. Taniguchi, E. Demler *et al.*, Kinetic magnetism in triangular moiré materials, *Nature (London)* **623**, 509 (2023).
- [32] J. Sung, J. Wang, I. Esterlis, P. A. Volkov, G. Scuri, Y. Zhou, E. Brutschea, T. Taniguchi, K. Watanabe, Y. Yang, M. A. Morales, S. Zhang, A. J. Millis, M. D. Lukin, P. Kim, E. Demler, and H. Park, An electronic microemulsion phase emerging from a quantum crystal-to-liquid transition, *Nat. Phys.* **21**, 437 (2025).
- [33] A. Lucas and K. C. Fong, Hydrodynamics of electrons in graphene, *J. Phys. Condens. Matter* **30**, 053001 (2018).
- [34] R. Xue, N. Maksimovic, P. E. Dolgirev, L.-Q. Xia, R. Kitagawa, A. Müller, F. Machado, D. R. Klein, D. MacNeill, K. Watanabe, T. Taniguchi, P. Jarillo-Herrero, M. D. Lukin, E. Demler, and A. Yacoby, Signatures of magnon hydrodynamics in an atomically-thin ferromagnet, [arXiv:2403.01057](https://arxiv.org/abs/2403.01057).
- [35] Y. Saito, T. Nojima, and Y. Iwasa, Highly crystalline 2D superconductors, *Nat. Rev. Mater.* **2**, 16094 (2016).

- [36] N. J. McLaughlin, C. Hu, M. Huang, S. Zhang, H. Lu, G. Q. Yan, H. Wang, Y. Tserkovnyak, N. Ni, and C. R. Du, Quantum imaging of magnetic phase transitions and spin fluctuations in intrinsic magnetic topological nanoflakes, *Nano Lett.* **22**, 5810 (2022).
- [37] T. I. Andersen, B. L. Dwyer, J. D. Sanchez-Yamagishi, J. F. Rodriguez-Nieva, K. Agarwal, K. Watanabe, T. Taniguchi, E. A. Demler, P. Kim, H. Park, and M. D. Lukin, Electron-phonon instability in graphene revealed by global and local noise probes, *Science* **364**, 154 (2019).
- [38] P. E. Dolgirev, S. Chatterjee, I. Esterlis, A. A. Zibrov, M. D. Lukin, N. Y. Yao, and E. Demler, Characterizing two-dimensional superconductivity via nanoscale noise magnetometry with single-spin qubits, *Phys. Rev. B* **105**, 024507 (2022).
- [39] S. Chatterjee, P. E. Dolgirev, I. Esterlis, A. A. Zibrov, M. D. Lukin, N. Y. Yao, and E. Demler, Single-spin qubit magnetic spectroscopy of two-dimensional superconductivity, *Phys. Rev. Res.* **4**, L012001 (2022).
- [40] Z. Liu, R. Gong, J. Kim, O. K. Diessel, Q. Xu, Z. Rehfuss, X. Du, G. He, A. Singh, Y. S. Eo, E. A. Henriksen, G. D. Gu, N. Y. Yao, F. Machado, S. Ran, S. Chatterjee, and C. Zu, Quantum noise spectroscopy of superconducting dynamics in thin film $\text{Bi}_2\text{Sr}_2\text{CaCu}_2\text{O}_{8+\delta}$, [arXiv:2502.04439](https://arxiv.org/abs/2502.04439).
- [41] J. B. Curtis, N. Maksimovic, N. R. Poniatowski, A. Yacoby, B. Halperin, P. Narang, and E. Demler, Probing the Berezinskii-Kosterlitz-Thouless vortex unbinding transition in two-dimensional superconductors using local noise magnetometry, *Phys. Rev. B* **110**, 144518 (2024).
- [42] A. T. Boothroyd, Practical aspects of neutron scattering, in *Principles of Neutron Scattering from Condensed Matter* (Oxford University Press, New York, 2020), pp. 343–404.
- [43] F. Casola, T. van der Sar, and A. Yacoby, Probing condensed matter physics with magnetometry based on nitrogen-vacancy centres in diamond, *Nat. Rev. Mater.* **3**, 17088 (2018).
- [44] J. Rovny, S. Gopalakrishnan, A. C. B. Jayich, P. Maletinsky, E. Demler, and N. P. de Leon, Nanoscale diamond quantum sensors for many-body physics, *Nat. Rev. Phys.* **6**, 753 (2024).
- [45] L. Thiel, Z. Wang, M. A. Tschudin, D. Rohner, I. Gutiérrez-Lezama, N. Ubrig, M. Gibertini, E. Giannini, A. F. Morpurgo, and P. Maletinsky, Probing magnetism in 2D materials at the nanoscale with single-spin microscopy, *Science* **364**, 973 (2019).
- [46] P. Bhattacharyya *et al.*, Imaging the meissner effect in hydride superconductors using quantum sensors, *Nature (London)* **627**, 73 (2024).
- [47] M. J. H. Ku, T. X. Zhou, Q. Li, Y. J. Shin, J. K. Shi, C. Burch, L. E. Anderson, A. T. Pierce, Y. Xie, A. Hamo, U. Vool, H. Zhang, F. Casola, T. Taniguchi, K. Watanabe, M. M. Fogler, P. Kim, A. Yacoby, and R. L. Walsworth, Imaging viscous flow of the Dirac fluid in graphene, *Nature (London)* **583**, 537 (2020).
- [48] S. Kolkowitz, A. Safira, A. A. High, R. C. Devlin, S. Choi, Q. P. Unterreithmeier, D. Patterson, A. S. Zibrov, V. E. Manucharyan, H. Park, and M. D. Lukin, Probing Johnson noise and ballistic transport in normal metals with a single-spin qubit, *Science* **347**, 1129 (2015).
- [49] C. Du, T. van der Sar, T. X. Zhou, P. Upadhyaya, F. Casola, H. Zhang, M. C. Onbasli, C. A. Ross, R. L. Walsworth, Y. Tserkovnyak, and A. Yacoby, Control and local measurement of the spin chemical potential in a magnetic insulator, *Science* **357**, 195 (2017).
- [50] J. Rovny, Z. Yuan, M. Fitzpatrick, A. I. Abdalla, L. Futamura, C. Fox, M. C. Cambria, S. Kolkowitz, and N. P. de Leon, Nanoscale covariance magnetometry with diamond quantum sensors, *Science* **378**, 1301 (2022).
- [51] K.-H. Cheng, Z. Kazi, J. Rovny, B. Zhang, L. Nassar, J. D. Thompson, and N. P. de Leon, Massively multiplexed nanoscale magnetometry with diamond quantum sensors, *Phys. Rev. X* **15**, 031014 (2025).
- [52] M. Cambria, S. Chand, and S. Kolkowitz, Scalable parallel measurement of individual nitrogen-vacancy centers, *Phys. Rev. X* **15**, 031015 (2025).
- [53] W. S. Huxter, F. Dalmagioni, and C. L. Degen, Multiplexed scanning microscopy with dual-qubit spin sensors, [arXiv:2407.19576](https://arxiv.org/abs/2407.19576).
- [54] M. Joliffe, V. Vorobyov, and J. Wrachtrup, Readout of strongly coupled NV center-pair spin states with deep neural networks, *Quantum Sci. Technol.* **10**, 045006 (2025).
- [55] J. J. Carmiggelt, I. Bertelli, R. W. Mulder, A. Teepe, M. Elyasi, B. G. Simon, G. E. W. Bauer, Y. M. Blanter, and T. van der Sar, Broadband microwave detection using electron spins in a hybrid diamond-magnet sensor chip, *Nat. Commun.* **14**, 490 (2023).
- [56] C. S. Wolfe, V. P. Bhallamudi, H. L. Wang, C. H. Du, S. Manuilov, A. J. Berger, R. Adur, F. Y. Yang, and P. C. Hammel, Off-resonant manipulation of spins in diamond via precessing magnetization of a proximal ferromagnet, *Phys. Rev. B* **89**, 180406 (2014).
- [57] E. Lee-Wong, R. Xue, F. Ye, A. Kreisel, T. van der Sar, A. Yacoby, and C. R. Du, Nanoscale detection of magnon excitations with variable wavevectors through a quantum spin sensor, *Nano Lett.* **20**, 3284 (2020).
- [58] B. A. McCullian, A. M. Thabt, B. A. Gray, A. L. Melendez, M. S. Wolf, V. L. Safonov, D. V. Pelekhov, V. P. Bhallamudi, M. R. Page, and P. C. Hammel, Broadband multi-magnon relaxometry using a quantum spin sensor for high frequency ferromagnetic dynamics sensing, *Nat. Commun.* **11**, 5229 (2020).
- [59] G. Q. Yan, S. Li, H. Lu, M. Huang, Y. Xiao, L. Wernert, J. A. Brock, E. E. Fullerton, H. Chen, H. Wang, and C. R. Du, Quantum sensing and imaging of spin-orbit-torque-driven spin dynamics in the non-collinear antiferromagnet Mn_3Sn , *Adv. Mater.* **34**, 2200327 (2022).
- [60] F. Machado, E. A. Demler, N. Y. Yao, and S. Chatterjee, Quantum noise spectroscopy of dynamical critical phenomena, *Phys. Rev. Lett.* **131**, 070801 (2023).
- [61] M. E. Ziffer *et al.*, Quantum noise spectroscopy of critical slowing down in an atomically thin magnet, [arXiv:2407.05614](https://arxiv.org/abs/2407.05614).
- [62] Y. Li, Z. Ding, C. Wang, H. Sun, Z. Chen, P. Wang, Y. Wang, M. Gong, H. Zeng, F. Shi, and J. Du, Critical fluctuation and noise spectra in two-dimensional Fe_3GeTe_2 magnets, [arXiv:2407.00647](https://arxiv.org/abs/2407.00647).
- [63] A. Ariyaratne, D. Bluvstein, B. A. Myers, and A. C. B. Jayich, Nanoscale electrical conductivity imaging using a

- nitrogen-vacancy center in diamond, *Nat. Commun.* **9**, 2406 (2018).
- [64] E. Bersin, M. Walsh, S. L. Mouradian, M. E. Trusheim, T. Schröder, and D. Englund, Individual control and readout of qubits in a sub-diffraction volume, *npj Quantum Inf.* **5**, 38 (2019).
- [65] R. Monge, T. Delord, and C. A. Meriles, Reversible optical data storage below the diffraction limit, *Nat. Nanotechnol.* **19**, 202 (2023).
- [66] W. Ji, Z. Liu, Y. Guo, Z. Hu, J. Zhou, S. Dai, Y. Chen, P. Yu, M. Wang, K. Xia, F. Shi, Y. Wang, and J. Du, Correlated sensing with a solid-state quantum multisensor system for atomic-scale structural analysis, *Nat. Photonics* **18**, 43 (2024).
- [67] T. Delord, R. Monge, and C. A. Meriles, Correlated spectroscopy of electric noise with color center clusters, *Nano Lett.* **24**, 6474 (2024).
- [68] D. M. Irber, F. Poggiali, F. Kong, M. Kieschnick, T. Lüthmann, D. Kwiatkowski, J. Meijer, J. Du, F. Shi, and F. Reinhard, Robust all-optical single-shot readout of nitrogen-vacancy centers in diamond, *Nat. Commun.* **12**, 532 (2021).
- [69] L. Orphal-Kobin, K. Unterguggenberger, T. Pregolato, N. Kemf, M. Matalla, R.-S. Unger, I. Ostermay, G. Pieplow, and T. Schröder, Optically coherent nitrogen-vacancy defect centers in diamond nanostructures, *Phys. Rev. X* **13**, 011042 (2023).
- [70] See Supplemental Material at <http://link.aps.org/supplemental/10.1103/7xlj-52xt> for experimental details and theoretical derivations of correlation, which includes Refs. [71–78].
- [71] S. Sangtawesin, B. L. Dwyer, S. Srinivasan, J. J. Allred, L. V. H. Rodgers, K. De Greve, A. Stacey, N. Dontschuk, K. M. O'Donnell, D. Hu, D. A. Evans, C. Jaye, D. A. Fischer, M. L. Markham, D. J. Twitchen, H. Park, M. D. Lukin, and N. P. de Leon, Origins of diamond surface noise probed by correlating single-spin measurements with surface spectroscopy, *Phys. Rev. X* **9**, 031052 (2019).
- [72] S. Pezzagna, B. Naydenov, F. Jelezko, J. Wrachtrup, and J. Meijer, Creation efficiency of nitrogen-vacancy centres in diamond, *New J. Phys.* **12**, 065017 (2010).
- [73] L. M. Pham, S. J. DeVience, F. Casola, I. Lovchinsky, A. O. Sushkov, E. Bersin, J. Lee, E. Urbach, P. Cappellaro, H. Park, A. Yacoby, M. Lukin, and R. L. Walsworth, Nmr technique for determining the depth of shallow nitrogen-vacancy centers in diamond, *Phys. Rev. B* **93**, 045425 (2016).
- [74] D. A. Hopper, H. J. Shulevitz, and L. C. Bassett, Spin readout techniques of the nitrogen-vacancy center in diamond, *Micromachines* **9**, 437 (2018).
- [75] P. Szańkowski, M. Trippenbach, and L. Cywiński, Spectroscopy of cross correlations of environmental noises with two qubits, *Phys. Rev. A* **94**, 012109 (2016).
- [76] P. Thibaudau and D. Beaujouan, Thermostatting the atomic spin dynamics from controlled demons, *Physica (Amsterdam)* **391A**, 1963 (2012).
- [77] L. Childress, M. V. Gurudev Dutt, J. M. Taylor, A. S. Zibrov, F. Jelezko, J. Wrachtrup, P. R. Hemmer, and M. D. Lukin, Coherent dynamics of coupled electron and nuclear spin qubits in diamond, *Science* **314**, 281 (2006).
- [78] M. H. Abobeih, J. Cramer, M. A. Bakker, N. Kalb, M. Markham, D. J. Twitchen, and T. H. Taminiau, One-second coherence for a single electron spin coupled to a multi-qubit nuclear-spin environment, *Nat. Commun.* **9**, 2552 (2018).
- [79] A. Batalov, V. Jacques, F. Kaiser, P. Siyushev, P. Neumann, L. J. Rogers, R. L. McMurtrie, N. B. Manson, F. Jelezko, and J. Wrachtrup, Low temperature studies of the excited-state structure of negatively charged nitrogen-vacancy color centers in diamond, *Phys. Rev. Lett.* **102**, 195506 (2009).
- [80] M. Gross and S. Haroche, Superradiance: An essay on the theory of collective spontaneous emission, *Phys. Rep.* **93**, 301 (1982).
- [81] S. F. Yelin, M. Koštrun, T. Wang, and M. Fleischhauer, Correlation in superradiance: A closed-form approach to cooperative effects, [arXiv:quant-ph/0509184](https://arxiv.org/abs/quant-ph/0509184).
- [82] O. Rubies-Bigorda and S. F. Yelin, Superradiance and subradiance in inverted atomic arrays, *Phys. Rev. A* **106**, 053717 (2022).
- [83] X. Li, J. Marino, D. E. Chang, and B. Flebus, A solid-state platform for cooperative quantum dynamics driven by correlated emission, *Phys. Rev. B* **111**, 064424 (2025).
- [84] M. Fukami, J. C. Marcks, D. R. Candido, L. R. Weiss, B. Soloway, S. E. Sullivan, N. Deegan, F. J. Heremans, M. E. Flatté, and D. D. Awschalom, Magnon-mediated qubit coupling determined via dissipation measurements, *Proc. Natl. Acad. Sci. U.S.A.* **121**, e2313754120 (2024).
- [85] P. Maletinsky, S. Hong, M. S. Grinolds, B. Hausmann, M. D. Lukin, R. L. Walsworth, M. Loncar, and A. Yacoby, A robust scanning diamond sensor for nanoscale imaging with single nitrogen-vacancy centres, *Nat. Nanotechnol.* **7**, 320 (2012).
- [86] G. Wang, Y.-X. Liu, J. M. Schloss, S. T. Alsld, D. A. Braje, and P. Cappellaro, Sensing of arbitrary-frequency fields using a quantum mixer, *Phys. Rev. X* **12**, 021061 (2022).
- [87] A. Laraoui, F. Dolde, C. Burk, F. Reinhard, J. Wrachtrup, and C. A. Meriles, High-resolution correlation spectroscopy of ^{13}C spins near a nitrogen-vacancy centre in diamond, *Nat. Commun.* **4**, 1651 (2013).
- [88] D. R. Glenn, D. B. Bucher, J. Lee, M. D. Lukin, H. Park, and R. L. Walsworth, High-resolution magnetic resonance spectroscopy using a solid-state spin sensor, *Nature (London)* **555**, 351 (2018).
- [89] C. L. Degen, F. Reinhard, and P. Cappellaro, Quantum sensing, *Rev. Mod. Phys.* **89**, 035002 (2017).
- [90] U. von Lüpke and *et al.*, Two-qubit spectroscopy of spatiotemporally correlated quantum noise in Superconducting qubits, *PRX Quantum* **1**, 010305 (2020).
- [91] L. Robledo, L. Childress, H. Bernien, B. Hensen, P. F. A. Alkemade, and R. Hanson, High-fidelity projective read-out of a solid-state spin quantum register, *Nature (London)* **477**, 574 (2011).
- [92] N. U. Köyliüoğlu *et al.*, Interaction-enhanced many-body covariance magnetometry for strongly interacting quantum sensors (to be published).
- [93] H. Hosseinabadi *et al.*, Theory of correlated spectroscopy with two NV centers (to be published).
- [94] Y. Zhang, R. Samajdar, and S. Gopalakrishnan, Nanoscale sensing of spatial correlations in nonequilibrium current noise, [arXiv:2404.15398](https://arxiv.org/abs/2404.15398).

- [95] K. Agarwal, R. Schmidt, B. Halperin, V. Oganesyan, G. Zaránd, M. D. Lukin, and E. Demler, Magnetic noise spectroscopy as a probe of local electronic correlations in two-dimensional systems, *Phys. Rev. B* **95**, 155107 (2017).
- [96] S. O. Demokritov, V. E. Demidov, O. Dzyapko, G. A. Melkov, A. A. Serga, B. Hillebrands, and A. N. Slavin, Bose–Einstein condensation of quasi-equilibrium magnons at room temperature under pumping, *Nature (London)* **443**, 430 (2006).
- [97] J. Rovny, S. Kolkowitz, and N. P. de Leon, Multi-qubit nanoscale sensing with entanglement as a resource, [arXiv: 2504.12533](https://arxiv.org/abs/2504.12533).
- [98] Xuan Hoang Le, Pavel Dolgirev, Piotr Put, Eric Peterson, Arjun Pillai, Alexander A. Zibrov, Eugene Demler, Hongkun Park, and Mikhail D. Lukin, Data for main text and supplemental material of the paper “Wideband Covariance Magnetometry below the Diffraction Limit”, Version v1, Zenodo, 2025, <https://zenodo.org/records/17343902>.

End Matter

Appendix: Resonantly assisted spin-to-charge conversion for readout—Despite high spin contrast [Fig. 1(d)], resonant readout in this Letter only achieves $\sigma_R = 15$ because spin mixing within the excited states limits the readout duration to a few microseconds [70,91]. To collect more photons per shot, we convert the NV spin state to charge state (NV^-/NV^0), which can be distinguished by the same resonant excitation as before, but for millisecond periods [70]. To achieve good spin selectivity for this conversion, we again utilize the resonant excitation [68]: when the NV is simultaneously illuminated with this laser and a high-power 660 nm light, only the $|0\rangle$ population undergoes two-photon ionization to NV^0 and yields low photon counts during later readout, while $|\pm 1\rangle$ remain in NV^- and thus are bright [Figs. 5(b) and 5(c)]. These steps form the RA-SCC protocol, used for readout in all correlation measurements in this Letter.

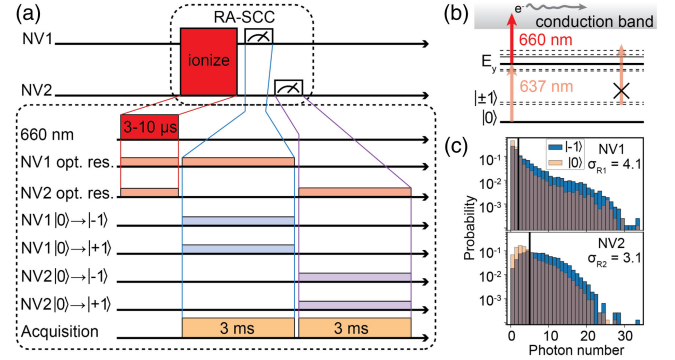


FIG. 5. RA-SCC protocol and readout noise characterization. (a) The protocol consists of spin-dependent ionization by a two-photon process shown in (b), followed by sequential readout of each NV charge state by resonant excitation. During this spin-agnostic readout, a low-power MW continuously mixes the spin states to counteract pumping into $|\pm 1\rangle$ [68]. (b) The resonant 637 nm laser selectively excites the $|0\rangle$ state, and a high-power 660 nm laser ionizes this population. (c) Photon statistics after RA-SCC for each NV, prepared in $|0\rangle$ or $|\pm 1\rangle$. Readout with lower (higher) photon counts than the threshold (bold vertical lines) are assigned to $|0\rangle$ ($|\pm 1\rangle$).